

Negation-forced amplitude amplification: fewer decisions, fewer gates, and ZX-guided condition selection for quantum constraint satisfaction

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Abstract

Dynamic quantum circuits make it tempting to solve constraint problems by measuring constraint ancillas mid-circuit and correcting violated constraints with classically controlled flips. We first prove that any such measure-and-flip circuit built from permutations, diagonal gates and computational-basis measurements is exactly a classical randomized algorithm, and confirm this numerically on the original proposals that motivated this work. We then show how the same intuition—*read each clause as the negation of a forbidden pattern and use it as a condition that forces a variable*—can be made genuinely quantum. In our *negation-forced* state preparation every variable is its own coin: it receives a Hadamard only if no clause forces it, so no decision register or history is needed. We prove that forced values never contradict a solution, which yields the exact success probability $p_\pi = \sum_{a \in \text{Sol}} 2^{-D_\pi(a)}$, where $D_\pi(a)$ counts unforced variables. Amplitude amplification over this preparation needs $O^*(2^{(1-1/k)n/2})$ gates for k -CNF, versus $O^*(2^{n/2})$ for Grover’s uniform preparation. Across 470 satisfiable instances of four families (random 3-SAT, 4-SAT, 1-in-3-SAT and graph 3-colouring, up to $n = 80$), the effective quantum cost base drops from 1.32^n – 1.38^n to 1.11^n – 1.20^n . Circuit-level simulations with matrix-product states (up to 245 qubits) reproduce the predicted probabilities. ZX-calculus simplification removes on average 51–70% of the T gates of each component, with verified equivalence, and makes the number of negative conditions worth checking an explicit, instance-dependent optimisation. Under a standard surface-code model these reductions lower the required code distance and physical-qubit count. We are explicit about the limits: the speedup is relative to Grover and to the classical algorithm with the same rules, not to modern CDCL solvers.

1 Introduction

Grover’s algorithm [1] and its generalisation, amplitude amplification [2], give a quadratic speedup for unstructured search. Applied to a constraint satisfaction prob-

lem (CSP) with n Boolean variables, Grover’s algorithm amplifies the uniform superposition and therefore needs $\Theta(\sqrt{2^n/M})$ oracle calls for M solutions—it ignores the structure of the constraints entirely.

Mid-circuit measurement and classical feed-forward (“dynamic circuits”) [3] suggest an appealing alternative: compute whether each constraint is violated into an ancilla, measure it, and flip variables when a constraint is violated. This paper started from such a proposal. Our first result (Section 3) is negative and general: when the measured quantities are functions of the computational-basis state and the corrections are permutations, the circuit is *exactly* a classical randomized algorithm, so no quantum advantage is possible regardless of qubit count.

The constructive part of the paper keeps the intuition and changes where it acts. A CNF clause $(\ell_1 \vee \dots \vee \ell_k)$ is the negation of the forbidden pattern $\neg \ell_1 \wedge \dots \wedge \neg \ell_k$. Read as a *negative condition*, it says: once all literals but one are false, the remaining one is forced. Using these conditions inside a coherent state preparation reduces the number of genuine decisions, which (i) raises the success probability exponentially, (ii) reduces the number of amplitude-amplification rounds quadratically in that gain, and therefore (iii) reduces the total gate count and the error-correction overhead.

Contributions.

1. A classicality lemma for measure-and-flip dynamic circuits, with numerical confirmation (Section 3).
2. An in-place *negation-forced* preparation A_π in which each variable is its own coin, a soundness lemma, an exact closed form for its success probability, and an $O^*(2^{(1-1/k)n/2})$ gate bound for amplitude amplification over it (Section 4).
3. A ZX-calculus co-design: verified T -count reductions of every component and a cost model that selects how many negative conditions to check (Section 5).
4. Large-scale numerics on four problem families up to $n = 80$, MPS circuit simulations up to 245 qubits, and fault-tolerant resource estimates (Section 6). All code and data are released.

2 Related work

Classical algorithms. The PPZ algorithm [4] assigns variables in random order, setting a variable when a clause forces it and guessing otherwise; its analysis shows that each variable is forced with probability at least $1/k$ for uniquely satisfiable k -CNF, giving $2^{(1-1/k)n}$ expected repetitions. PPSZ [5–7] strengthens the forcing rule with bounded-width resolution. Schönning’s random walk [8] instead repairs violated clauses.

Quantum speedups of randomized solvers. Any Las Vegas algorithm whose random choices can be put in superposition can be amplitude-amplified [2, 9]; this was used for Schönning and PPSZ-type algorithms [9, 10], and Montanaro’s quantum backtracking [11] speeds up tree search with propagation. Recent work places Grover inside local search [12]. Our exponent for full forcing coincides with amplified PPZ; our contributions are the in-place construction, the exact success formula, the condition-selection trade-off and the resource chain down to physical qubits.

Measurement-driven search. Search by measurement [13], damped and weakly measured search [14–16], measurement-driven SAT solvers [17, 18] and the quantum Lovász local lemma [19, 20] use measurements of constraint projectors. Our classicality lemma delineates when such schemes collapse to classical algorithms.

ZX-calculus. The ZX-calculus [21] with graph-theoretic simplification [22] reduces non-Clifford counts [23]; we use PyZX [24]. Relative-phase and measurement-based uncomputation tricks [25] are complementary.

3 Measure-and-flip dynamic circuits are classical

Lemma 1 (Classicality). *Consider a circuit on qubits initialised to $|0\rangle$, with a layer of Hadamards on a subset S , followed by any sequence of (i) permutation gates in the computational basis (X , $CNOT$, Toffoli, multi-controlled X), (ii) diagonal gates, (iii) computational-basis measurements and resets, and (iv) classically controlled versions of (i)–(iii). The joint distribution of all measurement outcomes equals that of the classical algorithm that samples the bits in S uniformly and applies (i), (iii), (iv) to bit strings.*

Proof. Let Δ be complete dephasing in the computational basis. For a permutation P and a diagonal unitary D , $\Delta(P\rho P^\dagger) = P\Delta(\rho)P^\dagger$ and $\Delta(D\rho D^\dagger) = \Delta(\rho)$; computational-basis measurement and reset commute with Δ , and so do their classically controlled versions. Hence inserting Δ right after the Hadamard layer does not

Table 1: Classicality check (Lemma 1): success probability of the measure-and-flip circuits (stabilizer simulation, 50 000 shots) versus the same rules executed classically (2×10^6 trials); all differences are within 2σ of shot noise.

Circuit	iters	quantum	classical
2-col path P10	2	0.0086	0.0078
2-col path P10	5	0.0624	0.0624
2-col path P10	10	1.0000	1.0000
2-col path P10	20	1.0000	1.0000
2-col path P30	7	0.0000	0.0000
2-col path P30	15	0.0001	0.0001
2-col path P30	30	1.0000	1.0000
2-col path P30	60	1.0000	1.0000
2-col path P60	15	0.0000	0.0000
2-col path P60	30	0.0000	0.0000
Subset sum $\{1, 2, 3, 4\}$, target 7, QFT adder: TV distance 0.0008			

change any outcome statistics. Since $\Delta(|+\rangle\langle+|) = \mathbb{I}/2$, the state after dephasing is a uniform mixture over bit strings on S , evolved by permutations and measurements, i.e. a classical randomized algorithm. $\square$

The lemma also applies at block level: a quantum adder acting on a register initialised to $|0\rangle$ is a permutation of basis states even if implemented with QFT rotations. Table 1 confirms the lemma on the two original proposals: a measure-and-flip 2-colouring of path graphs (simulated with the stabilizer method, n up to 60) and a measure-and-flip subset-sum circuit with a QFT adder.

4 Negation-forced state preparation

Let F be a CNF formula over $x_1, \dots, x_n$ and π an order of the variables.

Definition 1 (Negative conditions). *For a variable v , let $\mathcal{C}_1(v)$ ($\mathcal{C}_0(v)$) be the clauses containing the literal v ($\neg v$) whose other variables all precede v in π . Given an assignment of the preceding variables, v is forced to 1 (resp. 0) if some clause in $\mathcal{C}_1(v)$ ($\mathcal{C}_0(v)$) has all its other literals false. $\mathcal{C}^K$ keeps at most K clauses of each polarity.*

Definition 2 (In-place preparation A_π^K). *Process the variables in order π . For v : compute the flags f_1, f_0 (OR over clauses of the AND of their negated other literals), apply $CNOT(f_1 \rightarrow x_v)$, apply a Hadamard on x_v controlled on $\neg f_1 \wedge \neg f_0$, then uncompute the flags. All work qubits return to $|0\rangle$.*

The variable qubit is its own coin; no register of random choices and no history is stored. The circuit uses only Toffoli, controlled-Hadamard, CNOT and X gates (Algorithm 1); the number of Toffolis is $O(\sum_v |\mathcal{C}(v)| k) = O(km)$.

Algorithm 1 Negation-forced preparation A_π^K

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1: for  $v$  in order  $\pi$  do
2:   for  $c \in \mathcal{C}_1^K(v) \cup \mathcal{C}_0^K(v)$  do
3:      $a_c \leftarrow \bigwedge_{\ell \in c \setminus \{v\}} \neg \ell$  ▷ negative condition
4:   end for
5:    $f_1 \leftarrow \bigvee_{c \in \mathcal{C}_1} a_c$ ;  $f_0 \leftarrow \bigvee_{c \in \mathcal{C}_0} a_c$ 
6:    $x_v \oplus = f_1$ ; if  $\neg f_1 \wedge \neg f_0$  then  $H(x_v)$  ▷ decision
7:   uncompute  $f_1, f_0, a_c$ 
8: end for

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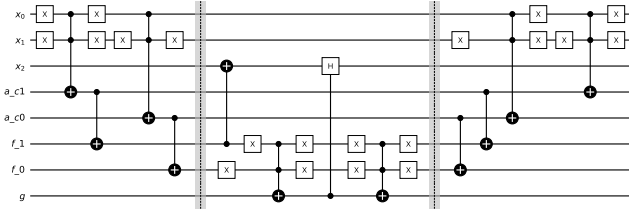


Figure 1: One negation-forced step for variable x_2 with clauses $(x_0 \vee x_1 \vee x_2) \in \mathcal{C}_1$ and $(\neg x_0 \vee \neg x_1 \vee \neg x_2) \in \mathcal{C}_0$. Left: negative conditions a_c (all other literals false) and flags f_1, f_0 . Middle: forced value and the decision (controlled Hadamard on the variable itself). Right: uncomputation; all ancillas return to $|0\rangle$ and are reused for the next variable.

Lemma 2 (Soundness). *If the variables preceding v agree with a satisfying assignment a , then a forced value of v equals a_v , and v is never forced both ways.*

Proof. If $c \in \mathcal{C}_1(v)$ has all other literals false under a , then a satisfies c only through the literal v , so $a_v = 1$. Symmetrically for $\mathcal{C}_0(v)$. Both cannot hold. $\square$

Lemma 3 (Exact success probability). *Let $D_\pi^K(a)$ be the number of unforced variables met when following a . Then*

$$p_\pi^K = \|\Pi_{\text{Sol}} A_\pi^K |0\rangle\|^2 = \sum_{a \in \text{Sol}} 2^{-D_\pi^K(a)}. \quad (1)$$

Moreover p_π^K is non-decreasing in K .

Proof. $A_\pi^K |0\rangle = \sum_x \alpha_x |x\rangle$ with $\alpha_x \geq 0$: each basis assignment x is produced by a unique branch, whose amplitude is $2^{-1/2}$ per Hadamard applied to $|0\rangle$ and 1 per forced step (all ancillas are restored). By Lemma 2 every solution a is reachable, its branch follows a , and $\alpha_a^2 = 2^{-D_\pi^K(a)}$. Increasing K enlarges the sets of forcing clauses, so $D_\pi^K(a)$ can only decrease. $\square$

Equation (1) reduces the computation of p_π to model enumeration and is what makes our large- n numerics exact. We verified it against exhaustive branch enumeration (zero discrepancy on all tested instances) and against circuit simulation (Section 6).

Theorem 1 (Gate complexity). *Let F be a k -CNF with n variables and m clauses and a unique solution. Amplitude amplification over A_π with random orders and*

an unknown-probability schedule [2, 26] finds the solution with $O(n 2^{((1-1/k)n+1)/2})$ expected applications of A_π , $A_\pi^\dagger$ and the phase oracle, i.e. $O^(2^{(1-1/k)n/2})$ gates, using $n + O(m)$ qubits.*

Proof. By Lemma 3, $p_\pi \geq 2^{-D_\pi(a^*)}$. The satisfiability coding lemma of PPZ [4] gives $\mu = \mathbb{E}_\pi[D_\pi(a^*)] \leq (1-1/k)n$. By Markov's inequality $\Pr_\pi[D_\pi(a^*) < \mu+1] \geq 1/(\mu+1)$, so after an expected $O(n)$ random orders one has $p_\pi \geq 2^{-(\mu+1)}$; for each order, amplitude amplification with an exponential schedule capped at $O(2^{(\mu+1)/2})$ rounds succeeds with constant probability [26]. Each application of A_π or the oracle costs $O(km)$ Toffolis. $\square$

For several solutions, Equation (1) sums over all of them and the bound only improves. The PPZ exponent is not new [9, 10]; what Lemma 3 adds is an exact, cheaply computable probability for *every* instance and every condition budget K , which enables the selection procedure of Section 5.

5 ZX-calculus co-design

Pipeline. Each component— A_π^K , the phase oracle O_F (clause ANDs, a global AND, a Z), and the reflection S_0 —is compiled to Clifford+ T , converted to a ZX-diagram, simplified with `full_reduce` and re-extracted [22, 24]. Compute/uncompute pairs of Toffolis that share controls fuse into phase gadgets, which is where most T gates disappear.

Verification. Every optimised component is checked against the original: exactly with statevectors for small instances, and for larger ones by running $U_{\text{orig}}^\dagger U_{\text{ZX}}$ on random inputs (with a Hadamard to expose relative phases) under MPS simulation and requiring that every shot returns the input.

Condition selection. Checking more negative conditions raises p_π^K (Lemma 3) but makes A_π^K more expensive. With $G(\cdot)$ the post-ZX T -count,

$$\mathcal{T}(K) = \frac{\pi}{4\sqrt{p_\pi^K}} (2G(A_\pi^K) + G(O_F) + G(S_0)), \quad K^* = \arg \min_K \mathcal{T}(K). \quad (2)$$

Both factors are computable without running the quantum algorithm: p_π^K via Equation (1) and G via ZX simplification.

6 Experiments

Setup. Families: random 3-SAT at clause density 4.2, random 4-SAT at 9.6, positive 1-in-3-SAT with 0.6 n triples (encoded as one 3-clause and three 2-clauses each), and Erdős-Rényi graph 3-colouring with two bits per vertex (average degree 4.2, or 3.0 below 10 vertices). Only satisfiable instances are kept; solutions are enumerated

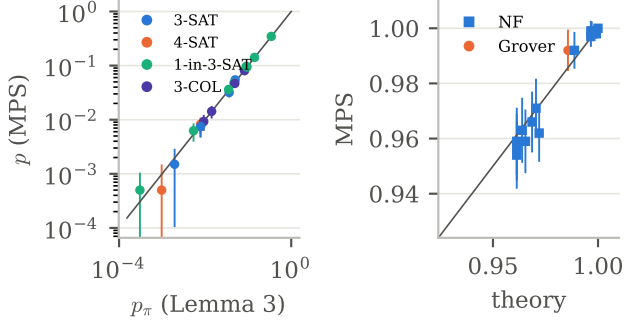


Figure 2: MPS circuit simulations versus theory. Left: success probability of $A_\pi|0\rangle$ (one point per instance; error bars are 2σ shot noise). Right: success after k^* amplification rounds for negation-forced (NF) and uniform (Grover) preparations; the label gives k^* .

with MiniSat [27] through PySAT [28] (instances with more than 2×10^4 solutions are rejected). Circuits are built in Qiskit [29]; MPS simulations [30] use Qiskit Aer; ZX optimisation uses PyZX [24]. All runs used two CPU cores.

E2: circuit-level validation with MPS. Figure 2 compares the success probability of the simulated circuit $A_\pi|0\rangle$ with Equation (1) for 19 instances with n up to 32 (up to 245 qubits), and the success probability after k^* full amplification rounds with the prediction $\sin^2((2k^* + 1)\arcsin\sqrt{p})$, for both the negation-forced preparation and Grover’s uniform preparation. 19 of 19 preparation points and 17 of 17 amplification points agree with theory within 3σ shot noise; the largest amplified circuit has 22070 gates on 205 qubits. Runs exceeding 15 minutes were stopped (preparation: 3-COL at $n = 20, 24$ and 3-SAT at $n = 24$; amplification: the uniform baseline for 4-SAT and beyond $n = 8$): the MPS bond dimension grows with n and the number of rounds, which limits the validation, not the algorithm.

E3: scaling across problem families. Figure 3 and Table 2 report $\log_2 p$ versus n for 470 instances (10 per size, 10 random orders each). Fitting $p \propto 2^{-sn}$ with bootstrap confidence intervals, the effective quantum base $2^{s/2}$ drops from 1.32^n – 1.38^n for the uniform preparation to 1.11^n – 1.20^n with negative conditions. The measured fraction of decisions per variable stays below the PPZ bound $1 - 1/k$ (Table 2).

E4: ZX simplification. Table 3 lists T and two-qubit gate counts per component before and after ZX simplification, together with the equivalence checks. On average ZX removes 51–70% of the T gates of each component, while the two-qubit count changes little; all 9 checked components passed the equivalence test. For 4-SAT, PyZX extraction of the oracle (77 four-literal clauses) did

Table 2: Fitted scaling $p \propto 2^{-sn}$ (95% bootstrap CI) and effective quantum base $2^{s/2}$; decisions per variable D/n along solutions versus the PPZ bound $1 - 1/k$ (for k -CNF).

Family	inst.	Grover	NF (median)	NF (best)	D/n (bound)
3-SAT	130	1.355	1.169 [1.161,1.177]	1.152	0.589 (0.667)
4-SAT	110	1.375	1.200 [1.191,1.208]	1.192	0.628 (0.750)
1-in-3-SAT	120	1.318	1.110 [1.103,1.117]	1.093	0.511 (–)
3-COL	110	1.357	1.187 [1.182,1.192]	1.181	0.630 (–)

Table 3: ZX simplification per component (mean over instances): T -count and two-qubit gates before $\rightarrow$ after, and the equivalence checks (statevector or MPS, see text).

Family	comp.	T	2q	ΔT
3-SAT	A_π	1338 $\rightarrow$ 536	1153 $\rightarrow$ 1325	–60%
3-SAT	O_F	2618 $\rightarrow$ 1130	2244 $\rightarrow$ 2991	–57%
3-SAT	S_0	182 $\rightarrow$ 56	157 $\rightarrow$ 117	–69%
1-in-3-SAT	A_π	473 $\rightarrow$ 205	448 $\rightarrow$ 555	–57%
1-in-3-SAT	O_F	1036 $\rightarrow$ 417	888 $\rightarrow$ 925	–60%
1-in-3-SAT	S_0	182 $\rightarrow$ 56	157 $\rightarrow$ 117	–69%
3-COL	A_π	1737 $\rightarrow$ 859	1506 $\rightarrow$ 2064	–51%
3-COL	O_F	2688 $\rightarrow$ 865	2304 $\rightarrow$ 2166	–68%
3-COL	S_0	238 $\rightarrow$ 72	205 $\rightarrow$ 130	–70%

Equivalence checks passed: 9/9 components.

not finish within one CPU-hour; ZX extraction cost is a practical limitation of the approach at this size.

E5: selecting negative conditions. Figure 4 shows $\mathcal{T}(K)$ from Equation (2) for $K \in \{0, 1, 2, 3, \infty\}$, where $K = 0$ is Grover’s algorithm. Over 12 instances the combined reduction of the expected T -count relative to Grover without ZX ranges from $2.8\times$ to $11.0\times$. The optimum is interior ($K^* < \infty$) for 10 of 12 instances (distribution of K^* : 1: 2, 2: 5, 3: 3, ∞ : 2), so checking every available negative condition is not always optimal; choosing K^* instead of $K = \infty$ saves up to $1.61\times$.

E6: fault-tolerant resources and classical baselines.

We count logical T gates of the generated circuits exactly for n up to 80, combine them with the median p from E3, and estimate surface-code resources [31, 32] with physical error rate 10^{-3} , threshold 10^{-2} , logical error $p_L(d) = 0.1(p/p_{th})^{(d+1)/2}$, a total failure budget of 1%, one 15-to-1 factory [33] and a $1\mu s$ code cycle; ZX ratios from E4 are applied as measured averages per component (for 4-SAT, the average over the other families). Table 4 reports the result. At the largest sizes, relative to Grover: 3-SAT ($n = 80$): T lower by 1.7×10^5 , d 41 $\rightarrow$ 29, $2.0\times$ fewer physical qubits; 4-SAT ($n = 70$): T lower by 1.2×10^4 , d 41 $\rightarrow$ 33, $1.5\times$ fewer physical qubits; 1-in-3-SAT ($n = 70$): T lower by 2.3×10^5 , d 35 $\rightarrow$ 23, $2.3\times$ fewer physical qubits; 3-COL ($n = 80$): T lower by 4.2×10^4 , d 41 $\rightarrow$ 31, $1.7\times$ fewer physical qubits.

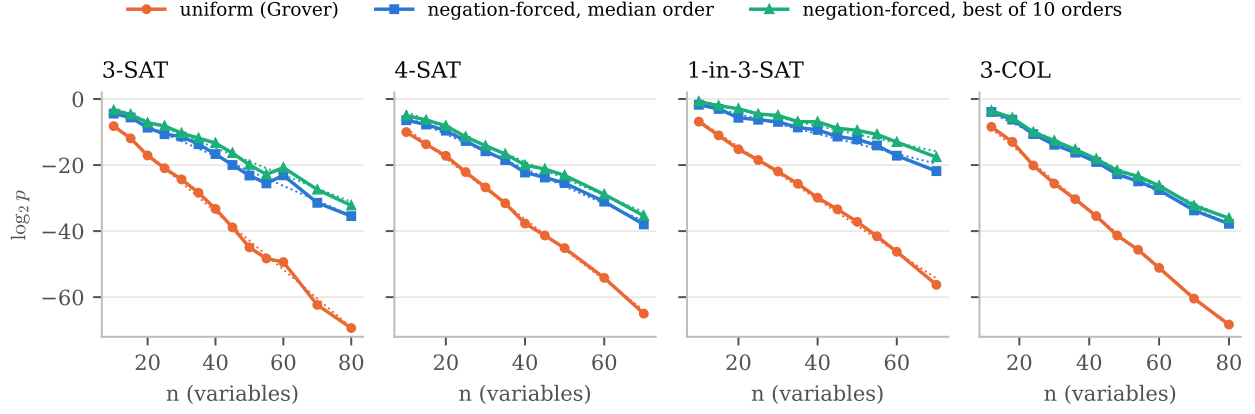


Figure 3: Success probability of the prepared state versus number of variables (median over 10 instances per size; dotted lines: least-squares fits). Orange: uniform superposition (Grover). Blue/green: negation-forced preparation, median and best of 10 variable orders.

Table 4: Fault-tolerant estimates for the median instance: expected logical T -count to find a solution, surface-code distance d , physical qubits and runtime (model in the text). Classical reps.: expected repetitions $1/p$ of the classical algorithm with the same negative conditions.

Family	n	Grover (uniform)				Negation-forced + ZX				classical reps.
		T	d	qubits	time	T	d	qubits	time	
3-SAT	40	1.0e9	29	9.7e5	8.2 h	2.5e6	23	6.1e5	58.1 s	1.0e5
3-SAT	80	5.3e14	41	3.8e6	690.3 y	3.1e9	29	1.9e6	1.0 d	3.9e10
4-SAT	40	1.4e10	31	2.3e6	5.1 d	6.4e7	27	1.8e6	28.8 min	4.8e6
4-SAT	70	3.2e14	41	7.2e6	420.4 y	2.6e10	33	4.6e6	10.0 d	2.6e11
1-in-3-SAT	40	1.4e8	25	4.4e5	58.7 min	8.3e4	19	2.5e5	1.6 s	6.5e2
1-in-3-SAT	70	2.3e12	35	1.5e6	2.6 y	1.0e7	23	6.4e5	3.8 min	3.3e6
3-COL	30	6.0e7	25	4.8e5	25.2 min	9.4e5	21	3.4e5	19.7 s	1.5e4
3-COL	80	4.2e14	41	3.4e6	552.7 y	1.0e10	31	1.9e6	3.6 d	2.4e11

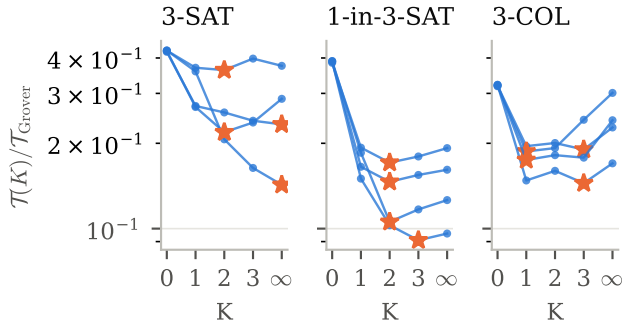


Figure 4: Total expected T -count (Equation (2), after ZX) versus the number of negative conditions K checked per variable and polarity, normalised by Grover ($K = 0$, no ZX). Each line is one instance; stars mark K^* .

7 Discussion and limitations

What the speedup is relative to. The gains are relative to Grover’s uniform preparation and are quadratic relative to the classical randomized algorithm that uses the same negative conditions (Table 4, column “classical

reps.”). They are *not* gains over industrial CDCL solvers: MiniSat solved every instance in our study with at most 43,965 conflicts. Random instances near the threshold are easy for CDCL at these sizes; our numerics measure exponents, not practical runtimes.

Qubits. Negative conditions do not reduce the number of logical qubits by themselves—the oracle’s clause register dominates. The physical-qubit savings come from fewer gates, which permits a smaller code distance.

“Partial” error tolerance. The algorithm is Las Vegas: candidate solutions are verified classically, so an error costs a repetition but never yields a wrong answer. This is error *detection* at the algorithmic level, not fault tolerance; we use it only to motivate the reduced error budget.

Scope of the numerics. ZX optimisation and full amplification were simulated only for small instances; the large- n resource estimates combine exact gate counts with ZX ratios measured at small n . The surface-code model is deliberately simple.

Outlook. Stronger negative conditions (PPSZ-style implications via bounded-width resolution) fit the same framework and Lemma 3 still applies whenever forced values are sound. Heralded weak measurements of the satisfaction flag can replace the unknown- p schedule on dynamic-circuit hardware.

8 Perspective: constraint-guided quantum computing

The results suggest a research programme that we tentatively call *constraint-guided quantum computing*: the co-design of the logical structure of a problem and the diagrammatic (ZX) structure of the circuit that searches it. Complexity-theoretic analyses typically treat the oracle and the initial state as black boxes of unit cost, while compilers optimise generic circuits without access to the logic of the problem; works such as quantum backtracking [11] and amplified classical solvers [9, 10] already connect the two at the level of query complexity, and the present work makes the connection explicit at the level of gates and fault-tolerant resources. We see four directions.

1. Informed instead of uniform preparation.

Search and variational algorithms usually start from $|+\rangle^{\otimes n}$. Constraint propagation, Boolean pruning and local consistency can be compiled into a coherent in-place preparation, so that amplification only explores assignments compatible with the problem’s logic; Lemma 3 shows that the benefit can be computed exactly beforehand.

2. Compilation for non-monotone trade-offs.

Checking every available condition is not always optimal (Figure 4): the right amount of logic to inject depends on the non-Clifford cost that survives ZX simplification. This calls for compilers that optimise jointly over the condition budget, the T -count and magic-state distillation.

3. Richer constraint languages.

Beyond CNF and graph colouring (and the knapsack example in the repository), the same soundness argument applies to general CSPs, pseudo-Boolean and mixed-integer constraints, routing and formal verification, where forced values take richer algebraic forms and compact encodings (adders, sorting networks) replace cover enumeration.

4. Hybridisation with classical reasoning engines.

Rather than competing with CDCL or CP-SAT solvers, a quantum routine can act as an amplification module inside a deductive pipeline, receiving the propagation rules (and, potentially, learned clauses) that the classical engine has already derived.

Whether these directions yield practical advantages remains open; the present results establish the exact ac-

counting needed to answer that question for each problem class.

9 Conclusion

Measure-and-flip constraint repair on dynamic circuits is classical. Used coherently, the same negative conditions shrink the space that amplitude amplification has to search: fewer decisions, exponentially higher success probability, fewer rounds and fewer gates, consistently across four problem families. ZX-calculus removes a large fraction of the remaining T gates and turns the choice of which conditions to check into a computable optimisation.

Code and data availability. The `negsearch` package, the experiment scripts, the raw results, the figure scripts, a hardware-ready notebook (multiple knapsack on IBM Quantum) and every problem instance used in this work (DIMACS format, with generator seeds and the variable orders evaluated) are available at <https://github.com/USERNAME/negsearch>.

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A Unknown success probability: heralded weak measurement

Theorem 1 uses an exponential schedule because p_π is unknown. On dynamic-circuit hardware an alternative is to copy the satisfaction flag weakly to a herald qubit inside every oracle call ($\text{CR}_y(2\varphi_t)$, no extra queries) and measure it before the reflection; a click projects onto the solution subspace and certifies the answer before readout [14–16]. With the schedule $\sin^2 \varphi_t = \min(1/2, 5/(t+1))$ the expected cost approaches $1.01\sqrt{N/M}$ without knowing M , versus $\approx 1.45\sqrt{N/M}$ for BBHT (Table 5); the 2-D model was validated against full circuit simulations. The same schedule applies verbatim to amplification over A_π with N/M replaced by $1/p_\pi$.

Table 5: Expected oracle queries to a verified solution, in units of $\sqrt{N/M}$ (2-D model, exact). Heralded: weak measurement of the satisfaction flag with $\sin^2 \varphi_t = \min(1/2, 5/(t+1))$, no knowledge of M .

$\log_2(N/M)$	Grover (M known)	BBHT	Heralded
6	0.88	1.60	0.78
10	0.81	1.63	1.00
14	0.79	1.52	1.02
18	0.79	1.48	1.01
22	0.79	1.44	1.01
24	0.79	1.43	1.01

B Noisy intermediate-scale simulation

For completeness, Table 6 reports a small instance under the **FakeFaz** noise model with relaxation charged to every mid-circuit measurement. At $N = 16$ all methods are at the noise floor (cost ≈ 8 , equal to guessing and verifying); measurement-based uncomputation (MBU: ancillas uncomputed by X -basis measurement and classically controlled phase fixes [25]), which needs many mid-circuit measurements, degrades sharply with latency. “Textbook” uses Qiskit’s multi-controlled gates; “coherent-rcx” uses relative-phase Toffolis; “Heralded T ” is the scheme of Appendix A with at most T iterations. The advantages discussed in the paper therefore concern query and gate complexity in the fault-tolerant regime, not present-day hardware.

Table 6: Noisy simulation with the **FakeFaz** (IBM Heron) noise model, binary 2×2 Sudoku ($N = 16$, $M = 2$); L : idle relaxation added per mid-circuit measurement. Cost: oracle queries per valid solution (random guessing: 8). Cert.: $P(\text{valid} \mid \text{herald click})$.

Method	L (ns)	$P(\text{valid})$	cert.	cost
Grover textbook	0	0.401	–	7.5
Grover coherent-rcx	0	0.358	–	8.4
Grover MBU-rcx	0	0.421	–	7.1
Heralded $T=2$	0	0.380	0.67	7.0
Heralded $T=4$	0	0.339	0.57	11.3
Heralded $T=6$	0	0.336	0.49	14.0
Grover textbook	2000	0.401	–	7.5
Grover coherent-rcx	2000	0.358	–	8.4
Grover MBU-rcx	2000	0.102	–	29.3
Heralded $T=2$	2000	0.340	0.60	7.9
Heralded $T=4$	2000	0.308	0.51	12.5
Heralded $T=6$	2000	0.297	0.43	16.1

References

[1] Lov K. Grover. A fast quantum mechanical algorithm for database search. In *Proceedings of the 28th*

Annual ACM Symposium on Theory of Computing, pages 212–219, 1996.

- [2] Gilles Brassard, Peter Høyer, Michele Mosca, and Alain Tapp. Quantum amplitude amplification and estimation. In *Quantum Computation and Information*, volume 305 of *Contemporary Mathematics*, pages 53–74. 2002.
- [3] Antonio D. Córcoles et al. Exploiting dynamic quantum circuits in a quantum algorithm with superconducting qubits. *Physical Review Letters*, 127:100501, 2021.
- [4] Ramamohan Paturi, Pavel Pudlák, and Francis Zane. Satisfiability coding lemma. *Chicago Journal of Theoretical Computer Science*, 1999(11), 1999.
- [5] Ramamohan Paturi, Pavel Pudlák, Michael E. Saks, and Francis Zane. An improved exponential-time algorithm for k-SAT. *Journal of the ACM*, 52(3):337–364, 2005.
- [6] Timon Hertli. 3-SAT faster and simpler—unique-SAT bounds for PPSZ hold in general. *SIAM Journal on Computing*, 43(2):718–729, 2014.
- [7] Dominik Scheder. PPSZ is better than you think. *TheoretCS*, 2024. FOCS 2021; arXiv:2207.11071.
- [8] Uwe Schöning. A probabilistic algorithm for k-SAT and constraint satisfaction problems. In *40th Annual Symposium on Foundations of Computer Science*, pages 410–414, 1999.
- [9] Andris Ambainis. Quantum search algorithms. *ACM SIGACT News*, 35(2):22–35, 2004.
- [10] Vedran Dunjko, Yimin Ge, and J. Ignacio Cirac. Computational speedups using small quantum devices. *Physical Review Letters*, 121:250501, 2018.
- [11] Ashley Montanaro. Quantum-walk speedup of backtracking algorithms. *Theory of Computing*, 14(15):1–24, 2018.
- [12] Yingyu Zhang. A hybrid quantum–classical algorithm for k-SAT via Grover-enhanced local search, 2026. Preprints.org 202609.1379.
- [13] Andrew M. Childs, Enrico Deotto, Edward Farhi, Jeffrey Goldstone, Sam Gutmann, and Andrew J. Landahl. Quantum search by measurement. *Physical Review A*, 66:032314, 2002.
- [14] Ari Mizel. Critically damped quantum search. *Physical Review Letters*, 102:150501, 2009.
- [15] Pablo Andrés-Martínez and Chris Heunen. Weakly measured while loops: peeking at quantum states, 2020. arXiv:2009.08832.

- [16] Yash Prabhat, Snigdha Thakur, and Ankur Raina. Ancilla-mediated fixed-point quantum search using Grover iterations, 2026. arXiv:2608.29839.
- [17] Simon C. Benjamin, Liming Zhao, and Joseph F. Fitzsimons. Measurement-driven quantum computing: Performance of a 3-SAT solver, 2017. arXiv:1711.02687.
- [18] Yipei Zhang, Philippe Lewalle, and K. Birgitta Whaley. Solving k-SAT problems with generalized quantum measurement, 2024. arXiv:2406.13611.
- [19] Martin Schwarz, Toby S. Cubitt, and Frank Verstraete. An information-theoretic proof of the constructive commutative quantum Lovász local lemma, 2013. arXiv:1311.6474.
- [20] András Gilyén and Or Sattath. On preparing ground states of gapped Hamiltonians: An efficient quantum Lovász local lemma. In *58th Annual Symposium on Foundations of Computer Science*, pages 439–450, 2017.
- [21] Bob Coecke and Ross Duncan. Interacting quantum observables: categorical algebra and diagrammatics. *New Journal of Physics*, 13:043016, 2011.
- [22] Ross Duncan, Aleks Kissinger, Simon Perdrix, and John van de Wetering. Graph-theoretic simplification of quantum circuits with the ZX-calculus. *Quantum*, 4:279, 2020.
- [23] Aleks Kissinger and John van de Wetering. Reducing the number of non-Clifford gates in quantum circuits. *Physical Review A*, 102:022406, 2020.
- [24] Aleks Kissinger and John van de Wetering. PyZX: Large scale automated diagrammatic reasoning. In *Proceedings of QPL 2019*, volume 318 of *EPTCS*, pages 229–241, 2020.
- [25] Craig Gidney. Halving the cost of quantum addition. *Quantum*, 2:74, 2018.
- [26] Michel Boyer, Gilles Brassard, Peter Høyer, and Alain Tapp. Tight bounds on quantum searching. *Fortschritte der Physik*, 46:493–505, 1998.
- [27] Niklas Eén and Niklas Sörensson. An extensible SAT-solver. In *Theory and Applications of Satisfiability Testing (SAT 2003)*, volume 2919 of *LNCS*, pages 502–518, 2004.
- [28] Alexey Ignatiev, Antonio Morgado, and Joao Marques-Silva. PySAT: A Python toolkit for prototyping with SAT oracles. In *SAT 2018*, volume 10929 of *LNCS*, pages 428–437, 2018.
- [29] Ali Javadi-Abhari et al. Quantum computing with Qiskit, 2024. arXiv:2405.08810.
- [30] Guifré Vidal. Efficient classical simulation of slightly entangled quantum computations. *Physical Review Letters*, 91:147902, 2003.
- [31] Austin G. Fowler, Matteo Mariantoni, John M. Martinis, and Andrew N. Cleland. Surface codes: Towards practical large-scale quantum computation. *Physical Review A*, 86:032324, 2012.
- [32] Daniel Litinski. A game of surface codes: Large-scale quantum computing with lattice surgery. *Quantum*, 3:128, 2019.
- [33] Sergey Bravyi and Alexei Kitaev. Universal quantum computation with ideal Clifford gates and noisy ancillas. *Physical Review A*, 71:022316, 2005.